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Key Points:
- An isentropic diagnostics gives the meridional distribution of stratosphere-troposphere exchange
- The meridional structure of cross-tropopause ozone flux has a strong connection with the jet
- Diabatic heating (mixing) contributes to an upward (downward) cross-tropopause ozone flux

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Quantifying isentropic stratosphere-troposphere exchange of ozone

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Abstract There is increased evidence that stratosphere-troposphere exchange (STE) of ozone has a significant impact on tropospheric chemistry and radiation. Traditional diagnostics of STE consider the ozone budget in the lowermost stratosphere (LMS) as a whole. However, this can only render the hemispherically integrated ozone flux and therefore does not distinguish the exchange of ozone into low latitudes from that into high latitudes. The exchange of ozone at different latitudes may have different tropospheric impacts. This present study extends the traditional approach from the entire LMS to individual isentropic layers in the LMS and therefore gives the meridional distribution of STE by the latitudes where each isentropic surface intersects the tropopause. The specified dynamics version of the Whole Atmosphere Community Climate Model is used to estimate the STE ozone flux on each isentropic surface. It is found that net troposphere-to-stratosphere ozone transport occurs in low latitudes along the 350–380 K isentropic surfaces and that net stratosphere-to-troposphere ozone transport takes place in the extratropics along the 280–350 K isentropes. Particularly, the seasonal cycle of extratropical STE ozone flux in the Northern Hemisphere displays a maximum in late spring and early summer, following the seasonal migration of the upper tropospheric jet and associated isentropic mixing. Furthermore, differential diabatic heating and isentropic mixing tend to induce STE ozone fluxes in opposite directions, but the net effect results in a spatiotemporal pattern similar to the STE ozone flux associated with isentropic mixing.

1. Introduction

There is substantial evidence for an acceleration of the Brewer-Dobson circulation (BDC) in the stratosphere in recent decades [see *Butchart et al.*, 2014, and references therein]. This acceleration in BDC leads to an increased downward ozone flux from the stratosphere to the troposphere, which is widely reported in climate models [e.g., *Hess et al.*, 2015] and is predicted to continue in the future [*Collins et al.*, 2003; *Hegglin and Shepherd*, 2009]. On the one hand, the increase of stratosphere-troposphere exchange (STE) of ozone will modify the tropospheric oxidative capacity [e.g.,*Kentarchos and Roelofs*, 2003], influencing the lifetime of methane [*Fiore et al.*, 2002] as well as the chemical cycles of other tropospheric pollutants. On the other hand, deep intrusion of stratospheric ozone, on some occasions, has been shown to greatly increase the surface ozone concentration and therefore becomes important for surface air quality prediction and control [*Lin et al.*, 2012].

Quantification of STE on the global or hemispheric scale can be inferred either from the residual circulation [*Holton*, 1990] or from the lowermost stratosphere (LMS) mass budget [*Appenzeller et al.*, 1996]. A number of studies found that the annual mean global ozone STE is within the range of 400–600 Tg/yr in both observations and chemistry transport models (CTMs) [e.g., *Gettelman et al.*, 1997; *Olsen et al.*, 2004; *Hsu and Prather*, 2005; *Stevenson et al.*, 2006; *Hsu and Prather*, 2009]. However, detailed spatial distribution of ozone STE is not well quantified, in spite of the fact that ozone STE over different regions may have different tropospheric impacts via different transport mechanisms. Exchange across the higher-latitude tropopause may transport ozone along the low isentropic surfaces that intersect the ground, resulting in a deep intrusion of stratospheric ozone into the planetary boundary layer (PBL) and with resultant air quality concerns. By contrast, ozone STE over lower latitudes limits its influence mostly to the middle and upper troposphere, and it is more likely to impact tropospheric oxidative capacity [*Kentarchos and Roelofs*, 2003]. In this study, an isentropic diagnostic of STE [*Nakamura*, 2007] is implemented to the LMS ozone budget simulated in a CTM. This diagnostic extends

Appenzeller et al.'s [1996] approach into individual isentropic layers in the LMS, such that the latitudes of STE can be differentiated by the intersection between an isentropic surface and the tropopause.

The seasonal cycle of STE ozone flux, particularly the timing of the maximum in STE, is also important, as the ozone STE may have a longer and more widespread influence in winter due to the longer photochemical time scale of ozone during the winter months [*Liang et al.*, 2009]. In the observations, there is a springtime maximum in the tropospheric column ozone in the Northern Hemisphere (NH) midlatitudes, particularly on the western edge of Europe [*Monks*, 2000]. This maximum may be explained partly by the large ozone STE during spring months [*Liang et al.*, 2009]. A springtime ozone maximum is consistent with the seasonality of stratospheric fallout of radioactive debris from nuclear bomb tests [e.g., *Fry et al.*, 1960] as well as with the spring maximum in the extratropical downwelling air mass flux in observations [e.g., *Appenzeller et al.*, 1996; *Gettelman et al.*, 1997] or idealized models [e.g., *Chen and Sun*, 2011]. However, trajectory analysis [*Wernli and Bourqui*, 2002; *James et al.*, 2003; *Skerlak et al.*, 2014] finds that deep stratospheric intrusions, which effectively transport ozone to the surface, maximize in winter. A springtime maximum also contrasts with various studies of tracer advection with reanalysis winds [*Chen*, 1995; *Haynes and Shuckburgh*, 2000; *Nakamura*, 2007], where the largest isentropic air mass transport is likely associated with monsoonal circulations during the summer months above the 340 K isentropic surface.

The widespread discrepancies in when and where ozone enters the troposphere may be attributed to the variety of mechanisms controlling STE and the lack of understanding about these mechanisms. It is well known that the BDC provides as a global constraint on the strength of STE flux of air mass. The acceleration of BDC simulated in many models [e.g., *Garcia and Randel*, 2008; *Yang et al.*, 2014] is consistent with the predicted increase of annual mean global ozone STE in a future climate. As to the temporal and spatial variability in the ozone STE, a number of studies have linked the occurrence of STE to the tropopause folding or cutoff cyclones [*Danielsen*, 1968; *Reed and Vicek*, 1969; *Holton et al.*, 1995], the summer monsoon or deep convection over continents [*Tang et al.*, 2011], and large-scale wave breaking over the Pacific and Atlantic oceans leading to isentropic mixing [*Chen*, 1995; *Postel and Hitchman*, 1999; *Jing et al.*, 2004; *Nakamura*, 2007].

Although numerous mechanisms for the STE have been analyzed, their relative contributions to the ozone STE variability have not been elucidated. *Schoeberl* [2004] divided the STE mass flux into a diabatic flux across isentropic surfaces, representing the slow but long-term transport by the BDC, and an adiabatic flux along isentropic surfaces, representing the more effective and transient exchange mostly due to isentropic mixing. However, limited by the framework of *Appenzeller et al.* [1996], *Schoeberl* [2004] can only partition the STE budget on the hemispheric scale. By contrast, the isentropic diagnostic of STE used in this study can partition the cross-tropopause mass flux along individual isentropic surface into two components that can be separately attributed to differential diabatic heating and isentropic mixing.

A 19 year simulation from the specified dynamics (SD) version of the Whole Atmosphere Community Climate Model (WACCM) is used in this study to examine the meridional distribution of STE ozone flux as well as to understand the mechanisms that lead to the STE. Section 2 introduces the isentropic diagnostic that quantifies the STE flux along individual isentropic surface and how to partition the STE flux into two components associated with diabatic heating and isentropic mixing, respectively. Details about the model (SD-WACCM) is introduced in section 3. Quantification of the spatiotemporal variability of ozone STE is given in section 4. Possible dynamical mechanisms for the diagnosed STE are discussed in section 5. Lastly, conclusions are drawn in section 6.

2. STE Diagnostics

2.1. Quantification of STE

Prior diagnostics of STE [*Appenzeller et al.*, 1996] have considered the air mass budget over the entire LMS. As illustrated in Figure 1a, the rate of change in air mass within the grey shaded LMS ($\partial m_{\mathrm{LMS}}/\partial t$) can be calculated as the difference between the air mass flux across the upper boundary of 380 K isentropic surface ($F_{380\mathrm{K}}$) and the flux across the lateral boundary of the tropopause, which is the targeted STE mass flux (F_{STE}). Using the notation that the flux from the troposphere to the stratosphere is positive, the STE mass flux following the mass continuity in the LMS reads

$$F_{\mathrm{STE}} = \frac{\partial m_{\mathrm{LMS}}}{\partial t} + F_{380\mathrm{K}} \tag{1}$$

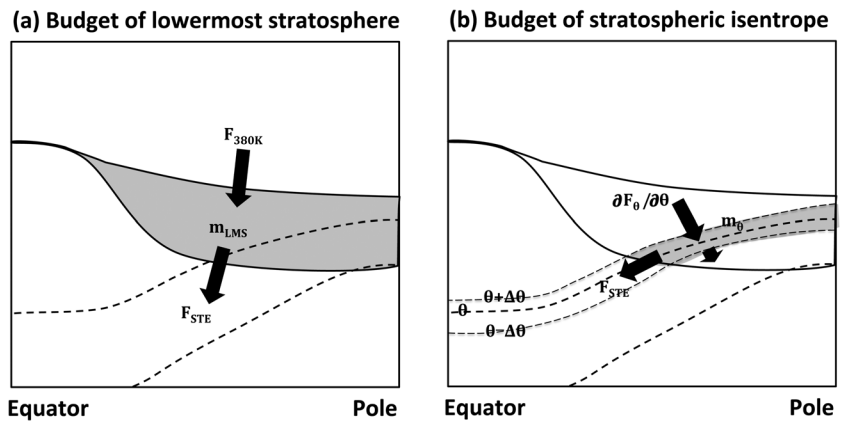

Figure 1. Schematics for the air mass budget (a) in the lowermost stratosphere (LMS), m_{LMS}, and (b) in an isentropic layer of the LMS, m_θ, which are both denoted by the grey shaded region. The cross-tropopause mass flux F_{STE} is calculated by the difference between the divergence of cross-isentropic mass flux (F_{380K} or F_θ) and the air mass tendency in the region of interest. Details are referred to the text in section 2.1.

where $F_{380K}=\langle\sigma\dot{\theta}\rangle_{380K}$, representing the hemispherically integrated diabatic air mass flux across the 380 K isentropic surface [*Schoeberl*, 2004]. Specifically, $\sigma=-g^{-1}\frac{\partial p}{\partial\theta}$ is the isentropic mass density, p is the pressure, θ is the potential temperature, $\dot{\theta}$ denotes the diabatic heating rate, and $<>$ denotes a hemispheric integral. Since the air mass fluxes across the 380 K isentropic surface and the tropopause are predominantly downward [e.g., *Appenzeller et al.*, 1996], as depicted in Figure 1, and all fluxes are upward positive by definition, the signs of F_{STE} and F_{380K} in equation (1) are negative.

In this LMS bulk methodology above, the quantification of STE is constrained on the hemispheric scale and therefore provides no information on the spatial distribution of STE. The latter is important so as to distinguish STE over high latitudes from its counterpart over low latitudes with distinct chemical and climate impacts as noted in the section 1.

By contrast, *Nakamura* [2007] divides the LMS into stacks of slim slabs enclosed by adjacent isentropic surfaces (see their Figure 3). In each isentropic slab (e.g.,see the grey shaded region in Figure 1b), we define an operator as the mass-weighted integral of the quantity C within the stratospheric region of this isentropic slab

$$M(C)=\int\int_{\text{stratospheric isentrope}}\sigma C\mathrm{d}S \tag{2}$$

For example, $F_\theta=M(\dot{\theta})$ denotes the total diabatic mass flux across the θ surface within the stratosphere. Hence, the air mass tendency within the isentropic slab $\left(\frac{\partial m_\theta}{\partial t}\right)$, following the mass continuity, is produced from the difference between the diabatic flux divergence $\left(\frac{\partial F_\theta}{\partial\theta}\right)$ in the direction perpendicular to the isentropic surfaces and the STE mass flux across the lateral boundary (F_{STE}).

$$F_{STE}=\frac{\partial m_\theta}{\partial t}+\frac{\partial F_\theta}{\partial\theta} \tag{3}$$

Note that equation (3) is mathematically written on an isentropic surface (different from the LMS methodology), so that all terms are expressed in the unit of kg sec^{-1} K^{-1}, and the diagnosed flux is isentropic. The discretization of the derivatives is shown in Appendix A, and it can physically induce flux that is not necessarily isentropic given that we are computing the STE flux as a residual. However, we argue that these errors are small, and the diagnosed STE flux is dominantly isentropic (not shown). The hemispherically integrated STE air mass flux in equation (1) can be recovered by integrating equation (3) up with all the isentropic slabs that intersect the tropopause from the pole to the equator.

This isentropic STE flux in each slab covers only a narrow meridional range on the tropopause between the two adjacent isentropic surfaces. Since the potential temperature increases monotonically with height and higher isentropic surfaces normally intersect with the tropopause at lower latitudes, a vertically aligned series of isentropic STE from high isentropic surfaces to low isentropic surfaces approximately renders the

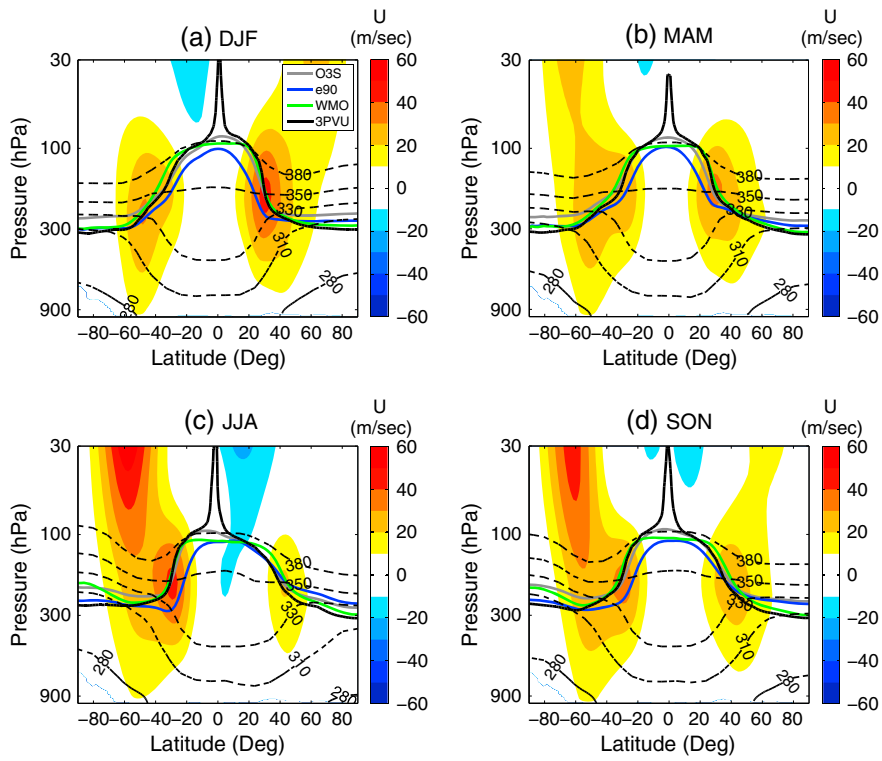

Figure 2. Latitude-altitude pattern of the tropopause in four seasons: (a) December–February, (b) March–May, (c) July–August, and (d) September–November. In each subplot, the tropopauses are shown as solid lines and are separately defined by (grey) stratospheric ozone O3S, (blue) idealized tracer e90, (green) WMO lapse rate criterion, and (black) 3 PVU. See section 3 for detailed definitions. The seasonally averaged zonal mean zonal winds (unit: m/s) are displayed in shading, and selected isentropic surfaces (K) are denoted as the dashed lines.

meridional distribution of STE from low latitudes to high latitudes. Note that although this isentropic STE diagnostic quantifies the meridional distribution of STE, the diagnosed STE fluxes do not map strictly onto the Eulerian latitudes but onto the tropopause-based "equivalent latitudes" instead (see details in *Nakamura* [1995]). Similar to *Nakamura* [2007], we divide the LMS by isentropic surfaces ranging from 280 K to 380 K, with a 4 K interval. The seasonal mean locations of selective isentropic surfaces are given as the dashed contours in Figure 2.

In addition, the chemical production/destruction of ozone should be accounted for in the ozone budget. By coupling the air mass budget in equation (3) with ozone mixing ratio, we have the STE ozone flux (F^{χ}_{STE}) as

$$F^{\chi}_{\mathrm{STE}} = \frac{\partial M(\chi)}{\partial t} + \frac{\partial M(\chi\dot{\theta})}{\partial \theta} - M(\dot{\chi}) \tag{4}$$

where χ is the mass mixing ratio of ozone in the unit of kg/kg, and $\dot{\chi}$ denotes the chemical sources (positive)/sinks (negative) of ozone.

2.2. Dynamic Partition of STE

Since potential vorticity (PV) is materially conserved in the absence of diabatic heating and friction, it is often used as a meridional coordinate that highlights the effect of nonconservative processes on mass transport. As the tropopause is often defined by a PV threshold, the cross-tropopause mass flux can be formulated using PV as the meridional coordinate such that the STE mass flux can be dynamically partitioned and attributed to different sources or sinks of PV.

More specifically, *Nakamura* [1995] showed that the isentropic STE mass flux, the mass flux across the PV isopleth Q, is mathematically equal to the PV sources/sinks ($\dot{q}$) averaged at the PV isopleth.

$$F_{\mathrm{STE}} = -\frac{\partial M(\dot{q})}{\partial q}\bigg|_{q=Q} \tag{5}$$

Furthermore, the PV sources/sinks can be separated into diabatic and frictional components [e.g.,*Holton*, 2004]:

$$\dot{q} = \frac{Dq}{Dt} \approx \dot{q}_s + \dot{q}_\kappa \tag{6}$$

Here $\dot{q}_s$ denotes the PV change associated with differential diabatic heating, which can be approximated as $\dot{q}_s \approx \frac{q}{\sigma}\frac{\partial(\sigma\dot{\theta})}{\partial\theta}$. $\dot{q}_\kappa$ denotes the PV change associated with frictional processes. If the frictional process is dominated by large-scale stirring and small-scale diffusion, it can be parameterized as $\dot{q}_\kappa = \kappa\nabla^2 q$, where κ is a diffusion coefficient [*Nakamura*, 2007]. Thus, we can examine the mechanisms influencing the STE by decomposing the STE mass flux in equation (3) into two components: the flux associated with differential diabatic heating (F_{dia} hereafter) and the other flux associated with isentropic mixing (F_{mix} hereafter). Such a decomposition is important because even with zero net STE mass flux, isentropic mixing can result in a net ozone flux along the cross-tropopause ozone gradient due to the two-way mass exchange.

$$F_{\mathrm{STE}} = F_{\mathrm{dia}} + F_{\mathrm{mix}} = -\left.\frac{\partial M(\dot{q}_s)}{\partial q}\right|_{q=Q} - \left.\frac{\partial M(\dot{q}_\kappa)}{\partial q}\right|_{q=Q} \tag{7}$$

Given the challenge in computing all the resolved, parameterized, and numerical frictional sinks of PV (including nudging), we use the model output of diabatic heating to compute F_{dia} directly and thus attain F_{mix} indirectly as a residue from $F_{\mathrm{mix}} = F_{\mathrm{STE}} - F_{\mathrm{dia}}$. To the extent that frictional processes are dominated by large-scale stirring and small-scale diffusion, the residue can be attributed to isentropic mixing, which can be approximated by the squared equivalent length ratio [*Haynes and Shuckburgh*, 2000].

Despite model differences, the numerical error induced by the nudging is assumed to be small [*Meloen et al.*, 2003], as the physical tendencies in the model are much less perturbed [*Jeuken et al.*, 1996]. Consequently, the diabatic heating is well constrained as a physical tendency, which lends more confidences to our estimate of F_{dia} and the dynamic partition followed. In addition, while the cross-isentropic diabatic flux F_θ and the isentropic mass flux due to diabatic PV source F_{dia} are both associated with diabatic processes, they represent the effect of diabatic heating in two different directions; F_θ denotes the mass transport perpendicular to the isentropic surface, whereas F_{dia} denotes the mass transport along the isentropic surface due to the diabatic source of PV.

For the STE ozone flux, equation (7) is rewritten as

$$F^{\chi}_{\mathrm{STE}} = F^{\chi}_{\mathrm{dia}} + F^{\chi}_{\mathrm{mix}} = -\left.\frac{\partial M(\chi\dot{q}_s)}{\partial q}\right|_{q=Q} - \left.\frac{\partial M(\chi\dot{q}_\kappa)}{\partial q}\right|_{q=Q} \tag{8}$$

3. SD-WACCM Simulation

The data analyzed here are from a control simulation of the specified dynamics version of the Whole Atmosphere Community Climate Model (SD-WACCM) from 1990 to 2009, nudged by the Modern-Era Retrospective Analysis for Research and Application (MERRA) reanalysis at every model time step [*Tang et al.*, 2013]. We discard the first year of 1990 to avoid any spin-up impacts and thus analyze a 19 year climatology of STE ozone flux. The SD-WACCM is a high-top climate-chemistry interactive model with an approximate horizontal resolution of 1.9° longitude by 2.5°latitude and vertically ranges from the surface to a top lid near 140 km. There are 88 unevenly spaced hybrid layers, and the vertical resolution near the tropopause is about 1 km. Details about the physical packages and nudging in the SD-WACCM are referred to *Tang et al.* [2013, and references therein]. The instantaneous meteorological and chemical variables are output at the frequency of once daily for offline analysis and is unlikely to be significantly affected by the diurnal cycle when the model output frequency has been increased to four times daily.

Before implementing the isentropic STE diagnostic, the model data on the hybrid coordinate are vertically interpolated onto the designated isentropic coordinate from 280 K to 380 K with an interval of 4 K, following the interpolation method described in *Edouard et al.* [1997]. This gives a reasonable isentropic representation of STE events simulated in the SD-WACCM, as the diagnosed STE flux displays almost no change with more isentropic levels included in the interpolation.

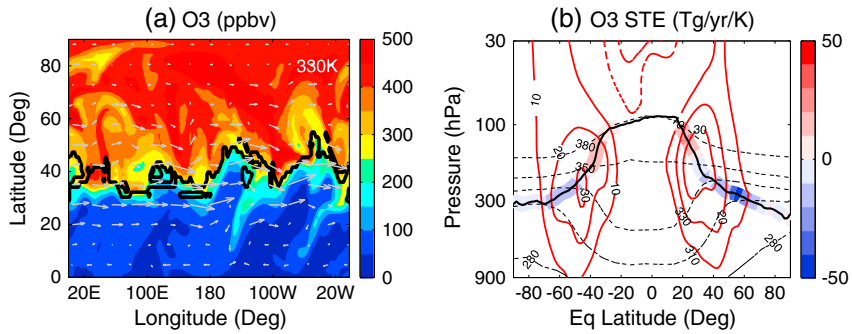

Figure 3. An ozone STE event at 0000 UTC 16 March 2001 in the SD-WACCM simulation. (a) Snapshot on the 330 K isentropic surface showing the longitude-latitude distribution of ozone mixing ratio (colors, unit: ppbv) and horizontal winds (vectors). The WMO tropopause is denoted as the bold solid black line. The ozone STE event is marked by the anticyclonic intrusion of high concentration of ozone across the tropopause over the eastern Pacific region (100°W–180°). (b) Meridional distribution of the diagnosed isentropic ozone flux (color blocks) across the the WMO tropopause (solid black line) as a function of equivalent latitude. Important isentropic surfaces (dashed lines) and zonal mean zonal wind (red contours) are marked similar to Figure 2.

As tropopause has a complicated structure, it is important to investigate the sensitivity of diagnosed STE flux to the choice of tropopause. Here we adopt four different types of tropopause definitions.

1. *O3S tropopause*. O3S denotes the ozone not produced by NO_x catalyzed tropospheric photochemical reactions [see *Hess and Zbinden*, 2013]. Here the O3S tropopause is defined by the value of O3S = 120ppbv.
2. *e90 tropopause*. e90 is an idealized tracer in the model, which has an *e*-folding decay time scale of 90 days. The e90 tropopause is defined by a threshold value of e90 concentration, at which 80% of the annually averaged atmospheric mass is below [*Prather et al.*, 2011].
3. *WMO tropopause*. The World Meteorological Organization (WMO) tropopause is defined using the lapse rate threshold of $\frac{dT}{dz} = -2\,\mathrm{K\,km^{-1}}$.
4. *PV tropopause*. Given the sharp PV gradient across the tropopause, the PV tropopause is defined by the threshold of 3 PVU in this study to be comparable with other tropopause definitions.

For the first three tropopause definitions, the tropopause is calculated online in the WACCM, and instantaneous tropopause pressure p^{trop} is saved daily for the off-line analysis. The PV tropopause, on the other hand, is defined by the PV calculated off-line as the absolute vorticity divided by isentropic static stability in the isentropic coordinates.

Figure 2 gives the locations of the tropopause as defined by four different definitions in the SD-WACCM simulation in each of four seasons. Overall, all tropopauses are spatially close to each other, and there is a similar seasonal cycle in terms of tropopause height for all of four tropopause definitions. However, there are also noticeable differences among these definitions. For example, the e90 tropopause is generally lower than others, particularly over the deep tropics.

4. Spatiotemporal Distribution of STE

In this section, we first examine a case study to illustrate the implementation of the method of diagnosing STE described above. The climatology of STE ozone flux, including its hemispheric budget and meridional distribution, is then given using the 19 year data set from the SD-WACCM.

4.1. Case Study

Hsu and Prather [2005] identified a strong STE event over the eastern Pacific in mid-March 2001 by considering the tropospheric ozone budget. Here we illustrate the isentropic diagnostic with this STE event based on the SD-WACCM simulation. Figure 3a gives the NH longitude-latitude map of the 330 K isentropic surface on the day of 16 March 2001 at 0000 UTC. The stratosphere is characterized by relatively higher concentration of ozone (red shading) poleward of the WMO tropopause (black solid contour), while the troposphere is demarcated by the lower ozone concentrations (blue shading). There is a noticeable anticyclonic wave breaking

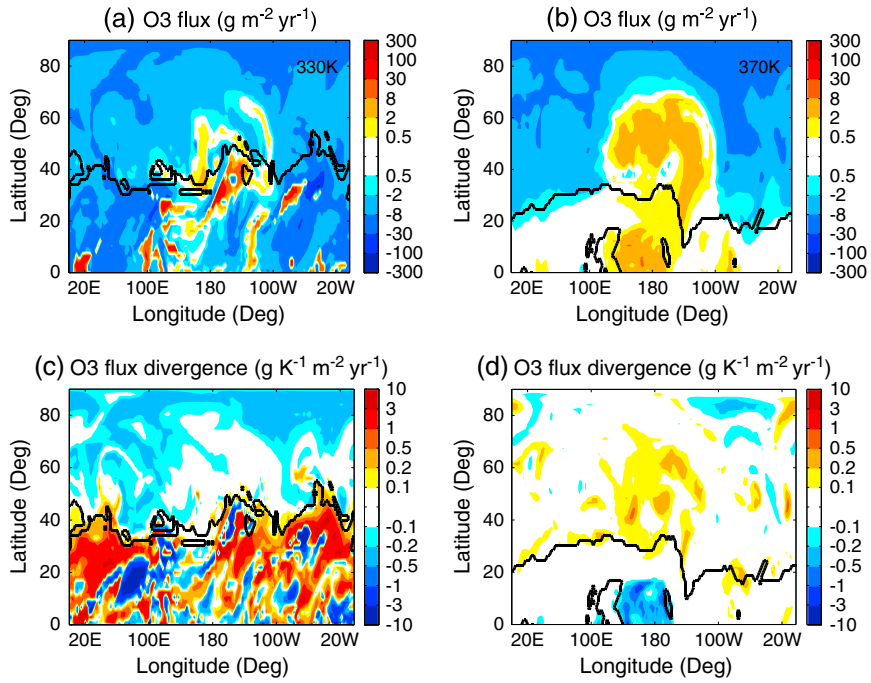

Figure 4. Examination on the isentropic STE diagnostic for the ozone STE event in Figure 3. Shadings denote (a and b) the diabatic ozone flux ($\chi\sigma\dot{\theta}$), and (c and d) the diabatic ozone flux divergence [$\partial(\chi\sigma\dot{\theta})/\partial\theta$] at the 330 K (Figures 4a and 4c) and 370 K (Figures 4b and 4d) isentropic surfaces. Other marks follow the same fashion as in Figure 3a. Details are referred to text in section 4.1.

feature near 140°W in the eastern Pacific Ocean, transporting ozone-abundant stratospheric air into the troposphere with the aid of strong equatorward winds (vectors) near 160°W.

Depicted in an equivalent latitude-pressure cross section, Figure 3b gives the corresponding isentropic ozone flux (colors) across the WMO tropopause (black solid contour) following equation (4). There is a clear spatial distribution in the diagnosed STE ozone flux, in terms of both isentropic surfaces (dashed lines) and equivalent latitudes. Specifically, net troposphere-to-stratosphere (upward hereafter) transport of ozone occurs on higher isentropic surfaces (lower equivalent latitudes) equatorward of the tropospheric jets (red contours), while net stratosphere-to-troposphere (downward hereafter) transport of ozone occurs on lower isentropes (higher equivalent latitudes) mostly poleward of the jets.

To further explain the upward (downward) STE ozone flux over high (low) isentropes in Figure 3b, the ozone budget in equation (4) is examined for two isentropic surfaces (i.e., 330 K and 370 K) in the NH. In a similar fashion as Figure 3a, the longitude-latitude distribution of diabatic ozone flux ($\chi\sigma\dot{\theta}$) across the 330 K and 370 K isentropic surface is given in Figures 4a and 4b, respectively. Both isentropic surfaces show a predominantly downward diabatic ozone flux in the stratosphere as expected from radiative cooling, except at 370 K where robust upwelling is centered at 180° longitude corresponding to the daytime radiative heating of stratospheric ozone. Interestingly, we find that the daily cross-tropopause ozone flux along the isentropes is largely balanced by the vertical divergence of diabatic ozone flux $\chi\sigma\dot{\theta}$ across the isentropes (not shown), although the change in the LMS ozone is also important. Therefore, on the 330 K isentropic surface (Figure 4c), the vertical convergence of diabatic ozone flux in the stratosphere (blue shading) indicates, by mass continuity, the isentropic transport of ozone from the stratosphere to the troposphere. In contrast, on the 370 K isentropic surface (Figure 4d), the vertical divergence of diabatic ozone flux in the stratosphere (red shading) implies that ozone is transported along the isentropic surface from the troposphere to the stratosphere.

4.2. Hemispherically Integrated STE Ozone Flux

We further evaluate the isentropic STE diagnostic developed here by comparing the 19 year climatology of hemispherically integrated STE ozone flux with those reported in previous studies using the LMS bulk methodology (see Figure 1a). Here the hemispherically integrated STE ozone flux is attained by integrating

Table 1. Hemispherically and Globally Integrated STE Ozone Fluxes F^{χ}_{STE} (Unit: Tg/yr) From the 19 Year SD-WACCM Simulation Using Four Different Tropopause Definitions (Columns 2–5)[a]

	F^{χ}_{STE} (Tg/yr)				Decomposition	
	O3S	e90	WMO	3 PVU	F^{χ}_{dia} (Tg/yr)	F^{χ}_{mix} (Tg/yr)
NH	−240 ± 20	−241 ± 19	−242 ± 19	−256 (−276) ± 20	713 ± 20	−969 ± 32
SH	−178 ± 20	−190 ± 19	−176 ± 19	−191 (−204) ± 19	681 ± 18	−872 ± 31
Global	−419 ± 36	−431 ± 35	−419 ± 34	−448 (−480) ± 35	1394 ± 34	−1841 ± 56

[a]The number in the parenthesis in column 5 show the results derived from the LMS bulk methodology, similar to equation (1). Dynamic decomposition of the STE ozone flux across the 3 PVU tropopause (column 5), into an ozone flux associated with the diabatic heating (F^{χ}_{dia}) and the other flux associated with the isentropic mixing (F^{χ}_{mix}), are shown in the last two columns following $F^{\chi}_{STE} = F^{\chi}_{dia} + F^{\chi}_{mix}$. Errors (after the ±) are given by the one standard deviation of the 19 year time series. See section 4.2 for details.

the isentropic STE fluxes from the 280 K isentropic surface to the 380 K isentropic surface, as described in section 2.1.

Table 1 summarizes the 19 year climatology of hemispherically and globally integrated STE ozone fluxes using the four tropopause definitions (see section 3). The climatological mean global ozone STE is in the range of 410–450 Tg/yr and is generally consistent with the reported numbers in many other studies using the LMS bulk methodology [e.g., *Gettelman et al.*, 1997; *Hegglin and Shepherd*, 2009]. A direct calculation following the LMS bulk methodology is additionally applied to the STE ozone flux across the 3 PVU tropopause (see Table 1). A marginally negligible difference is found between the two approaches, and the slight difference may mostly be contributed by the LMS discretization with limited numbers of isentropic layers. The ozone STE in the NH is 50–60 Tg/yr larger than its counterpart in the Southern Hemisphere (SH), but the STE air-mass flux between the two hemispheres are comparable (not shown). This implies that the smaller SH STE ozone flux may be dominantly attributed to the SH ozone variability, which the strong ozone depletion can significantly diminish the downward STE ozone flux maximum in the austral springtime. In addition, the hemispherically or globally integrated ozone STE flux in the annual mean is relatively insensitive to the choice of tropopause definition.

As to temporal evolution, the climatological seasonal cycle of hemispherically integrated ozone STE in the NH and in the SH is given in Figures 5a and 5b, respectively. The STE ozone flux is downward in both hemispheres throughout the year. In the NH, there is a noticeable peak in downward ozone STE in late spring and early summer, which is also consistent with the timing documented in many other studies using the LMS bulk methodology [e.g., *Olsen et al.*, 2004; *Hegglin and Shepherd*, 2009]. In contrast, relatively little seasonality is found in the SH, which both the SH ozone variability and less varied residual circulation (slightly weaker seasonality in the SH STE air-mass flux, not shown) can contribute. The seasonality of hemispherically integrated ozone STE is not sensitive to the choice of tropopause definition either, although the e90 tropopause exhibits a larger seasonality in the NH that can be attributed to the lower tropopause height in comparison

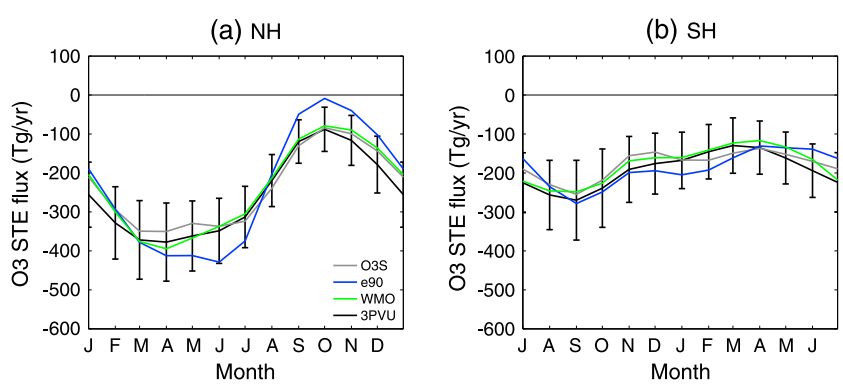

Figure 5. Seasonal cycle of the hemispherically integrated STE ozone fluxes (unit: Tg/yr) using four different tropopause definitions (color coded with the same fashion as in Figure 2) in the (a) Northern Hemisphere (NH) and (b) Southern Hemisphere (SH). Error bars are depicted as the one standard deviation deduced from the 19 year time series. Note that the abscissa in the SH is shifted forward by 6 months, and negative values indicate stratosphere-to-troposphere fluxes.

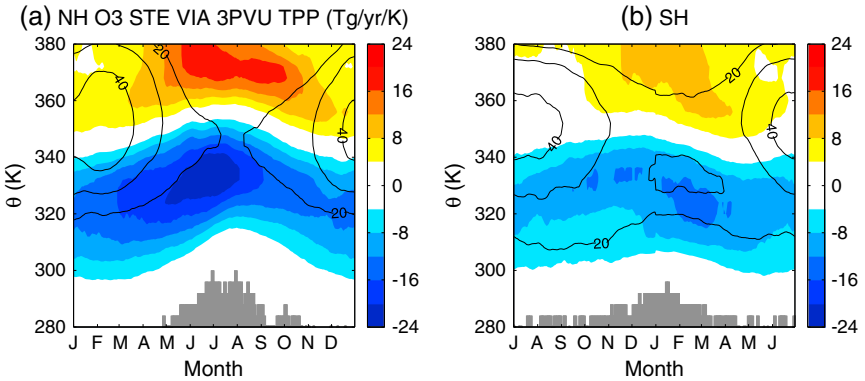

Figure 6. Seasonal cycle of the isentropic STE ozone fluxes (positive/negative values indicate troposphere-to-stratosphere/stratosphere-to-troposphere transport) across the 3 PVU tropopause (shading, unit: Tg/yr/K) and the zonal mean zonal wind on this tropopause (contours, unit: m/s) in (a) the NH and (b) the SH. Grey shaded regions identify the isentropic surfaces that do not intersect with the tropopause. The STE flux is calculated from daily data and smoothed by a 7 day running mean. The zonal mean zonal wind on the tropopause is obtained through an interpolation of the zonal mean zonal wind in the isentropic coordinates from a laitudinal series ϕ into a value affiliated with the designated tropopause-corresponding equivalent latitude ϕ_{eq}^{trop}.

with other tropopause definitions (Figure 2). This relative insensitivity to tropopause definition also applies to the remaining discussion of this paper. Therefore, we will focus on the 3 PVU tropopause hereafter.

4.3. Meridional Distribution of Ozone STE

As noted in section 2.1, one important merit of the isentropic diagnostic used in this paper is that it gives the meridional distribution of STE ozone flux, with exchange along the higher isentropic surfaces corresponding to lower latitude exchange and exchange along the lower isentropic surfaces corresponding to higher latitude exchange.

Figure 6 gives the climatological seasonal cycle of the isentropic STE ozone flux from the 280 K isentropic surface to the 380 K isentropic surface corresponding to a meridional ozone STE distribution from the poles to the low latitudes. Upward STE ozone fluxes (red shading) are found in low latitudes along the higher isentropic surfaces, whereas downward STE ozone fluxes (blue shading) are found in high latitudes along the lower isentropic surfaces. The boundary separating these two zones is situated near the core of the tropospheric jets (contours). While the maximum downward STE ozone flux shifts toward higher isentropic surfaces from winter to summer, the latitude where an isentropic surface intersects the tropopause also migrates seasonally (see Figure 2). Therefore, a migration of maximum STE ozone flux toward higher isentropic surface does not necessarily mean an equatorward shift in its latitudinal location. In addition, the maximum downward STE flux is found on the poleward flank of tropospheric jets, and it moves with the seasonal migration of the respective tropospheric jet. This indicates an intimate connection between the variability in ozone STE and the variability in tropospheric jet, which will be further discussed in section 5. Moreover, we note that the NH ozone STE (Figure 6a) is generally larger than its SH counterpart (Figure 6b) on almost all isentropic surfaces throughout the year. Again, the meridional distribution of isentropic STE ozone flux described above is relatively insensitive to the choice of tropopause definition (not shown).

To explicitly interpret the seasonal variability in ozone STE over different meridional regions and to relate this to their distinct impacts on tropospheric chemistry and surface air quality, we further aggregate the isentropic STE ozone fluxes into four regional bins as indicated by the dashed isentropic surfaces in Figure 2: (i) the jet's equatorward flank between 350 and 380 K, (ii) the jet core region between 330 and 350 K, (iii) the jet's poleward flank between 310 and 330 K, and (iv) the polar region between 280 and 310 K.

The STE ozone flux in each region is obtained by integrating over all the isentropic STE ozone fluxes within the region of interest. The seasonal cycle of STE ozone flux for the four regions is given in Figure 7. Consistent with the findings in Figure 6, the jet's equatorward flank (350–380 K) is characterized by a prominent upward STE ozone flux, whereas the downward STE ozone fluxes prevail in all other regions. In the NH, the upward STE ozone flux on the jet's equatorward side maximizes in summer, while the maximum downward STE ozone flux

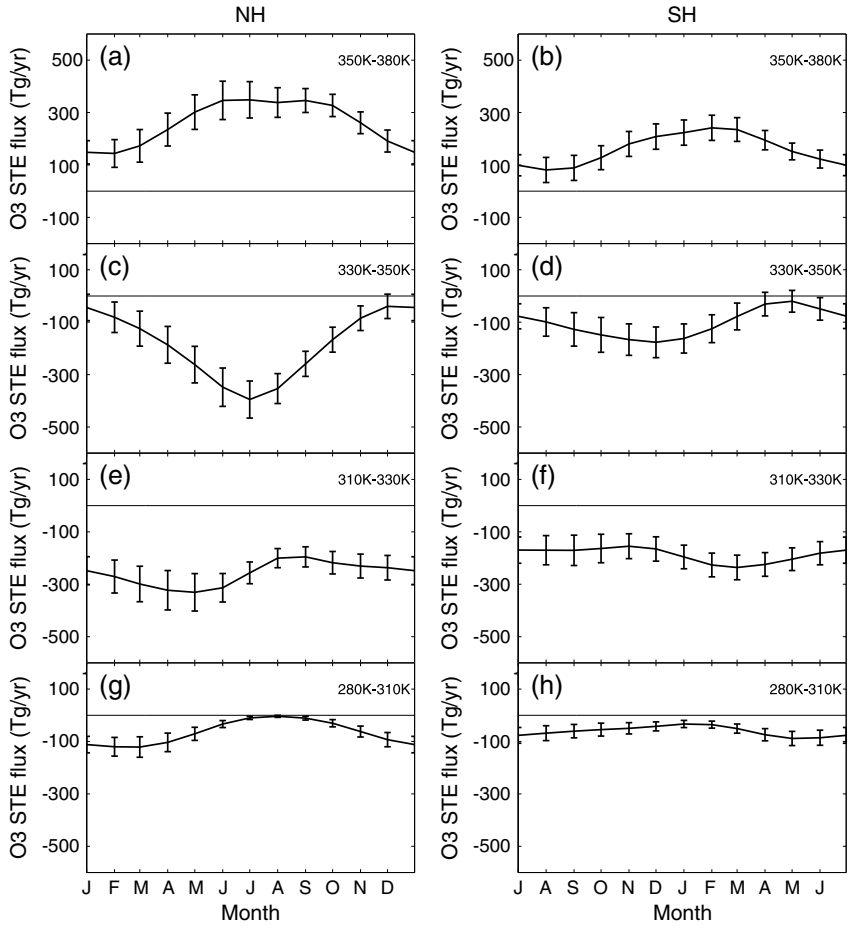

Figure 7. Similar to Figure 5, but for the STE ozone fluxes across the 3 PVU tropopause (a and b) on the jet's equatorward flank between the 350 and 380 K isentropic surfaces, (c and d) in the jet core region between 330 and 350 K, (e and f) on the jet's poleward flank between 310 and 330 K, and (g and h) in the polar region between 280 and 310 K. See details in section 4.3.

gradually shifts from higher isentropic surfaces in summer to lower isentropic surfaces in winter. The seasonal variability of STE ozone flux is less in the SH.

More importantly, from Figure 7, we propose that the discrepancies with respect to the seasonality of ozone STE reported in previous studies (see section 1) can be, at least partly, attributed to the different meridional regions of exchange analyzed in different papers. On the hemispheric scale, a late spring to early summer maximum (see Figure 5) has been widely reported in the NH [e.g., *Gettelman et al.*, 1997; *Hsu and Prather*, 2005]. This maximum can be mostly attributed to the downward STE ozone flux over the region near the poleward flank of the NH tropospheric jet (310K–330 K) that also peaks in the late spring (see Figure 7e). It is interesting to note that there is a large cancelation between the upward ozone STE on the jet's equatorward flank (350–380 K) and the downward ozone STE over the jet core region (330–350 K), and thus, the net effect contributes relatively less to the hemispherically integrated STE ozone flux. This cancelation is also probably indicative of recirculating ozone-rich air across the tropopause in regions equatorward of the tropospheric jet core. The summertime maxima of STE ozone flux above the 330 K isentrope is likely to be attributed to the monsoonal circulations (or deep convection over continents) [*Hsu and Prather*, 2005; *Tang et al.*, 2011] and associated large-scale isentropic mixing [*Chen*, 1995; *Nakamura*, 2007]. Furthemore, the wintertime maximum in ozone STE on the lower isentropic surfaces (280–310 K, and see Figure 7g) corresponds to the deep intrusion of stratospheric ozone reported in *James et al.* [2003] and *Skerlak et al.* [2014]. This downward STE ozone flux occurs on isentropic surfaces that intersect the surface of the Earth and thus possibly impacts surface air quality.

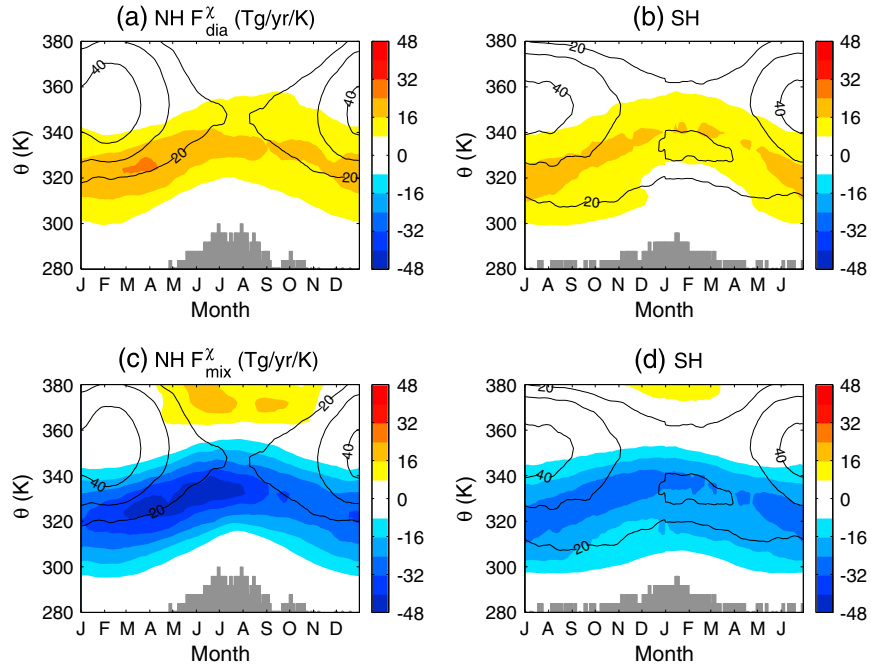

Figure 8. Similar to Figure 6 but for a dynamic partition of STE ozone flux across the 3 PVU tropopause; the net STE ozone flux (F^{χ}_{STE}) in Figure 6 is decomposed into (a and b) the STE flux associated with diabatic PV source (F^{χ}_{dia}) and (c and d) the STE flux associated with isentropic PV mixing (F^{χ}_{mix}).

5. Mechanisms of STE

In order to understand why the STE of ozone exhibits such a spatial distribution, we partition the STE flux into a diffusive term and and a diabatic term (see section 2.2). By doing this, we relate the diagnosed STE flux to diabatic and frictional processes that result in PV sources/sinks and hence to the mechanisms that control the spatiotemporal behaviors of STE ozone flux.

5.1. Dynamic Partition

Using equation (8), we can separate the STE ozone flux into the flux associated with diabatic PV source F^{χ}_{dia} and the flux associated with isentropic PV mixing F^{χ}_{mix}. For the climatological mean annual ozone budget, Table 1 (last two columns) shows that both components are about 3–4 times larger in magnitude than the net STE ozone flux. F^{χ}_{dia} contributes to a large upward STE ozone flux, while F^{χ}_{mix} contributes to a large downward ozone STE flux. There is a large cancelation between the two, but the F^{χ}_{mix} is slightly larger than the F^{χ}_{dia}, yielding the net downward STE ozone flux (F^{χ}_{STE}).

The spatial distribution of partitioned ozone STE fluxes is further illustrated in Figure 8. F^{χ}_{dia} is upward on all isentropic surfaces, while the F^{χ}_{mix} is predominantly downward on all isentropic surfaces in the extratropics. Both components maximize at the poleward edge of the tropospheric jets (contours) and the maxima shift seasonally with the jets. More importantly, the large cancelation between F^{χ}_{dia} and F^{χ}_{mix} on the hemispheric scale (see Table 1) occurs on almost all isentropic surfaces. This also sheds light on the difficulty in estimating the net STE magnitude since it is the small residue of the cancelation between two large terms. Because F^{χ}_{mix} is slightly larger than F^{χ}_{dia}, the net STE ozone flux exhibits a spatiotemporal pattern similar to the ozone flux associated with isentropic mixing (cf. Figures 8c and 8d and Figure 6).

5.2. Isentropic PV Mixing

Physically, isentropic mixing exchanges air masses across the tropopause in both directions, leading to transport of stratospheric ozone into the troposphere even with zero net mass flux. To explicitly document how the isentropic eddy mixing impacts the spatiotemporal variability of isentropic ozone STE, the extent of isentropic mixing is independently quantified by the concept of equivalent length based solely on the distortion of PV [*Nakamura*, 1995; *Haynes and Shuckburgh*, 2000].

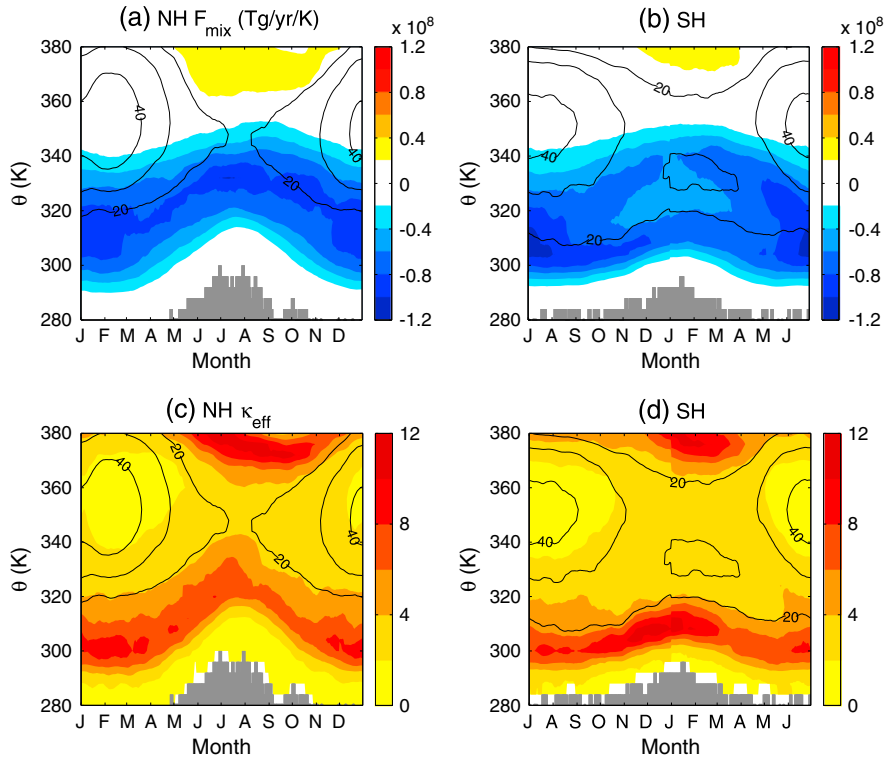

Figure 9. Similar to Figure 6 but for (a and b) the STE air mass flux associated with isentropic PV mixing (F_{mix}, unit: Tg/yr/K) and (c and d) the isentropic mixing represented by the squared equivalent length ratio of PV ($\tilde{\kappa}_{eff}$, dimensionless).

Figure 9 shows a clear connection between the STE air mass flux associated with isentropic PV mixing F_{mix} and the squared equivalent length ratio of PV $\tilde{\kappa}_{eff}$. The squared equivalent length ratio [*Allen and Nakamura*, 2001; *Haynes and Shuckburgh*, 2000] is a dimensionless measure of extent of isentropic mixing, and it is large where wave breaking takes place, and thus, the PV contours have complex geometry [*Haynes and Shuckburgh*, 2000]. In general, strong F_{mix} occurs over the region where wave breaking occurs frequently, indicated by the high $\tilde{\kappa}_{eff}$. In particular, Rossby waves prefer to break on the flanks of tropospheric jets, causing cross-tropopause air mass flux through PV streams or cutoff cyclones. In contrast, within the jet cores near the 350 K isentropic surface, isentropic mixing is largely suppressed due to the strong zonal winds and resultant reduced wave breaking, and a relatively small STE air mass flux is induced in the regions.

Additionally, the zonal wind, isentropic mixing, and F_{mix} show a similar seasonal variation. As the tropospheric jet tends to weaken from winter to summer, its poleward flank shifts toward higher isentropic surfaces. Consequently, the maximum of downward F_{mix} simultaneously shifts toward higher isentropic surfaces, following the movement of the jet edge and associated strong isentropic mixing (high $\tilde{\kappa}_{eff}$). Note that the maximum of air mass flux associated with isentropic mixing F_{mix} at 310–330 K is equatorward of the maximum of $\tilde{\kappa}_{eff}$ at 300–320 K throughout the year. This is because the latitudinal distribution of eddy PV fluxes (and associated mass fluxes by the zonal momentum constraint) depends on not only isentropic mixing but also the mean PV gradient that maximizes at the jet core [*Kunz et al.*, 2011].

As ozone is affected by chemical production/destruction as well as transport, the seasonal cycle of STE ozone flux is compared with that of ozone mixing ratio at the tropopause in Figure 10. If the ozone mixing ratio does not change in time, the STE ozone flux F^{χ}_{mix} is expected to vary seasonally in phase with the STE mass flux F_{mix}. Although in terms of hemispheric budget, the seasonal cycle of STE ozone flux F^{χ}_{mix} (Figures 10c and 10d, solid) resembles that of the mass flux F^{mix} (Figures 10a and 10b, dashed), the ozone flux lags the mass flux by a few months. This lag is likely due to the seasonal variability in ozone mixing ratio at the tropopause (Figures 10e and 10f, dash-dotted line), which maximizes in May in the NH [*Thouret et al.*, 2006] and is later

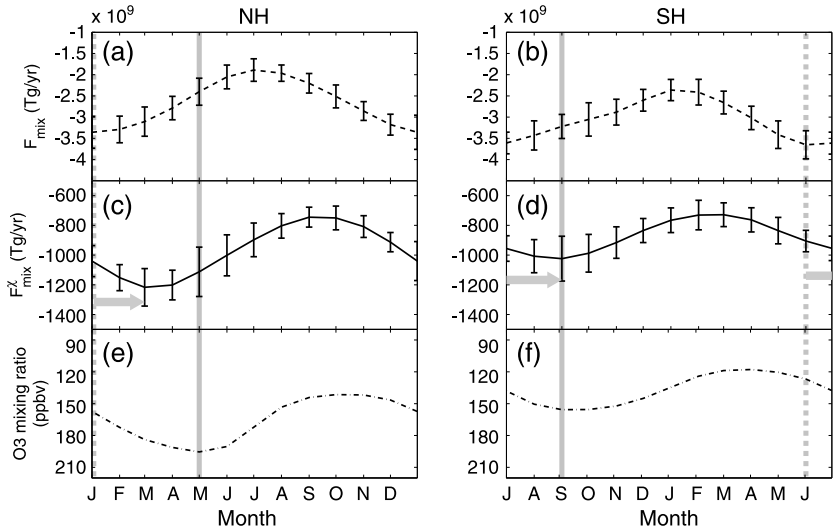

Figure 10. Similar to Figure 5 but for (a and b) the hemispherically integrated STE mass flux associated with isentropic PV mixing F_{mix} (dashed, unit: Tg/yr), (c and d) the STE ozone flux associated with isentropic PV mixing F^{χ}_{mix} (solid, unit: Tg/yr), and (e and f) the ozone mixing ratio at the 3 PVU tropopause χ (dash-dotted line, unit: ppbv). Vertically aligned grey dashed lines denote the month with maximum F_{mix}, and grey solid lines denote the month with maximum χ. Grey arrows denote the temporal shift of F_{mix} maximum to F^{χ}_{mix} maximum, due to the seasonal cycle of χ. Note that the ordinate for ozone mixing ratio is reversed.

than the cross-tropopause air mass flux F_{mix} that maximizes in January. It is likely that both the ozone chemical production/destruction and transport in the stratosphere can contribute to this seasonal variation in ozone mixing ratio at the tropopause. The relative importance of individual processes is beyond the scope of this study.

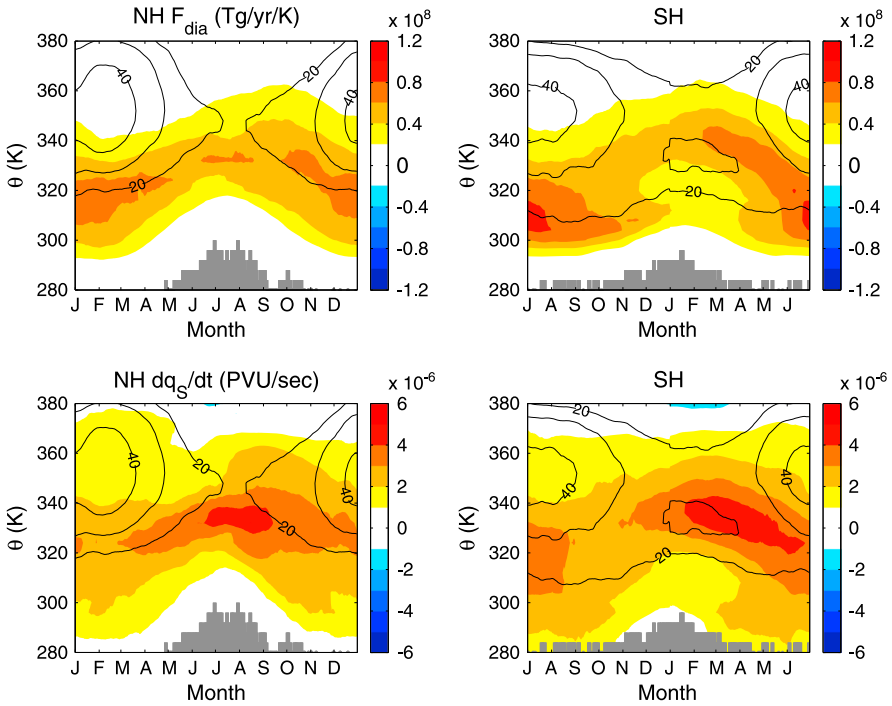

Figure 11. Similar to Figure 9, but for (a and b) the STE mass flux associated with diabatic PV source (F_{dia}, unit: Tg/yr/K), and (c and d) the diabatic PV source ($\dot{q}_S$, unit: PVU/s) on the 3 PVU tropopause. The derivation of $\dot{q}_S$ on the tropopause is similar to the approach described in Figure 6. Note that the sign of $\dot{q}_S$ has been flipped in the SH to account for the different sign of PV gradient across the tropopause.

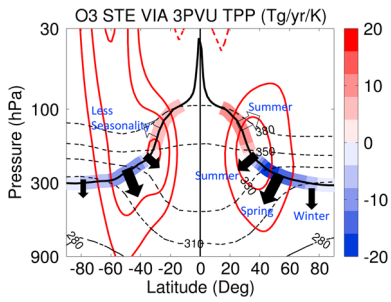

Figure 12. Schematic summary for the direction, magnitude, and seasonality of STE ozone flux over different isentropic (meridional) regions. Plotting follows the same fashion as in Figure 3b. Net stratosphere-to-troposphere ozone fluxes are denoted as filled arrows, and net troposphere-to-stratosphere ozone fluxes are denoted as open arrows. The size of arrow approximately denotes the magnitude of STE ozone flux.

5.3. Diabatic PV Source

To further interpret the diagnosed upward STE flux associated with the diabatic PV source, we use the fact that the STE air mass flux associated with diabatic PV source F_{dia} is approximately proportional to the mass-weighted average of diabatic PV source ($\dot{q}_S$) at the tropopause.

$$F_{\mathrm{dia}} = -\left.\frac{\partial M(\dot{q}_s)}{\partial q}\right|_{q=Q} = -\left(\frac{\partial q}{\partial m_\theta}\right)^{-1}\left.\frac{\partial M(\dot{q}_s)}{\partial m_\theta}\right|_{q=Q} \tag{9}$$

here the chain rule of derivative is used. Therefore, F_{dia} can be approximately related to the diabatic PV source ($\dot{q}_S$) at the tropopause.

Figure 11 shows the isentropic STE air mass flux associated with the diabatic PV source (F_{dia}, top) and the corresponding diabatic PV source ($\dot{q}_S$, bottom) averaged at the PV contour $Q = 3$ PVU. Note that the sign of $\dot{q}_S$ has been flipped in the SH to account for the different sign of PV gradient across the tropopause. Clearly, the upward F_{dia} can be understood as a decrease in tropopause height due to the differential diabatic heating (i.e., $Dq/Dt = \dot{q}_S > 0$ at the tropopause in the NH), engulfing the "original" tropospheric air into the expanding stratosphere. Interestingly, the seasonal variation of both F_{dia} and $\dot{q}_S$ exhibits an arch-type shape, similar to the seasonal movement of the jet's poleward flank (contours). Additionally, the seasonality in STE ozone flux associated with diabatic PV source F^{χ}_{dia} is affected by the seasonal cycle of ozone mixing ratio χ at the tropopause (not shown here).

6. Conclusions and Discussion

An isentropic diagnostic of STE has been applied to a 19 year simulation of the specified dynamics (SD) version of the Whole Atmosphere Community Model (WACCM), nudged by the MERRA reanalysis data. The diagnostic extends the hemispheric ozone budget in the lowermost stratosphere (LMS) to the budget over discretized isentropic layers. Given that low (high) isentropic surfaces intersect the tropopause at high (low) latitudes, the STE ozone fluxes along different isentropic surfaces render the meridional distribution of STE.

The overall findings are summarized in Figure 12. The climatological mean STE ozone flux is characterized by different direction, magnitude, and seasonality on different isentropic surfaces. We also find that the diagnosed ozone STE is relatively insensitive to different tropopause definitions, although the isentropic STE decreases in magnitude if the tropopause is defined at a much higher altitude (not shown here).

1. *Direction*. Net troposphere-to-stratosphere (upward) transport of ozone is found in the low latitudes, while net stratosphere-to-troposphere (downward) transport of ozone is found in the extratropics. The boundary separating these two regions lies at the core of the tropospheric jets.
2. *Magnitude*. The maxima in the extratropical downward STE of ozone lie on the poleward edge of the tropospheric jets, and their locations migrate seasonally with the jets.
3. *Seasonality*. The seasonal maximum of the STE ozone flux gradually migrates from the higher isentropic surfaces in summer to the lower isentropic surfaces in winter. The summertime maximum on the higher isentropic surfaces is likely associated with the monsoonal circulation (or deep convection over continents) [*Tang et al.*, 2011], while the wintertime maximum on the lower isentropic surfaces facilitates the deep intrusion of stratospheric ozone into the planetary boundary layer (PBL) [*Skerlak et al.*, 2014]. A pronounced STE maximum occurs in the NH in late spring and early summer, while much less seasonality is found in the SH.

To interpret the diagnosed spatiotemporal behaviors of STE ozone flux, a dynamic partition, embedded within the isentropic STE diagnostic, decomposes the net STE ozone flux into two components associated with either differential diabatic heating or isentropic PV mixing [*Nakamura*, 2007]. It is shown that the diagnosed spatiotemporal behaviors of ozone STE results from a large cancelation between the predominantly upward ozone STE flux associated with the diabatic PV source and the predominantly downward flux associated with

the isentropic PV mixing. As the latter is slightly larger than the former on almost all isentropic surfaces, the net STE ozone flux spatially and temporally resembles the STE ozone flux associated with isentropic PV mixing. Hence, the seasonality of the STE ozone flux can be thought of as the result of isentropic mixing of ozone across the tropopause. Furthermore, differential diabatic heating can alter the PV structure at the tropopause and induce the upward STE ozone flux by driving the tropopause downward so that tropospheric air is brought into the expanding stratosphere.

Given that the ozone mixing ratio decreases sharply from the stratosphere to the troposphere, the upward STE ozone flux associated with isentropic PV mixing (see Figures 8c and 8d) suggests an up-gradient transport of ozone from the troposphere to the stratosphere above the 350 K isentrope. Such an upward/poleward mass transport is also observed in the 350–370 K isentropic layer of the ERA-Interim Re-Analysis by *Homeyer and Bowman* [2013, Figure 6]. This may be explained by the asymmetry of wave breaking during summer monsoons, and the persistent cyclonic wave breaking can lead to preferred transport of air mass and ozone from the troposphere to the stratosphere.

Appendix A: Discretization of the Isentropic STE Diagnostic

Following equation (4), the isentropic STE diagnostic at the time step τ and isentropic layer k are discretized as

$$F^{\chi}_{\mathrm{STE}}|_{\tau,k} = \frac{M(\chi)|_{\tau+1,k} - M(\chi)|_{\tau-1,k}}{t|_{\tau+1} - t|_{\tau-1}} + \frac{M\left(\chi\dot{\theta}\right)\Big|_{\tau,k+1} - M\left(\chi\dot{\theta}\right)\Big|_{\tau,k-1}}{\theta|_{k+1} - \theta|_{k-1}} - M(\dot{\chi})|_{\tau,k} \tag{A1}$$

where the mass-weighted integral of the quantity C in the stratosphere $M(C)$ in equation (2) is discretely defined as

$$M(C)|_{\tau,k} = \sum_{j=j^*}\sum_{i=i^*} C_{j,i}\sigma_{j,i}\mathrm{dS}_{j,i}|_{\tau,k} \tag{A2}$$

The pairs of j^* (latitude) and i^* (longitude) identify the points in the stratosphere by either comparing with the tropopause pressure, p^{trop}

$$p_{j^*,i^*}|_{\tau,k} \le p^{\mathrm{trop}}_{j^*,i^*}|_{\tau,k} \tag{A3}$$

or the PV threshold $Q = 3$ PVU

$$\begin{aligned} q_{j^*,i^*}|_{\tau,k} &\ge Q_{j^*,i^*}|_{\tau,k} \ \text{ in the NH} \\ q_{j^*,i^*}|_{\tau,k} &\le Q_{j^*,i^*}|_{\tau,k} \ \text{ in the SH} \end{aligned} \tag{A4}$$

Here σ denotes the isentropic density, and $\mathrm{dS}_{j,i}$ denotes the area for the grid element at latitudinal point j and longitudinal point i, and adjacent half-grid points are used for the calculation of $\mathrm{dS}_{j,i}$, which $\mathrm{dS}_{j,i} = a^2\left(\sin\phi_{j+1/2} - \sin\phi_{j-1/2}\right)\left(\lambda_{i+1/2} - \lambda_{i-1/2}\right)$. ϕ is latitude, λ is longitude, and a is the planetary radius of the Earth. Note that the isentropic STE flux is calculated separately for each hemisphere.

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